

Assignment 1

Instructions

- PHY410: Do problems 1 and 2a-2b
- PHY 505: Do problems 1 and 2a-2c

Accept the assignment from “Classroom 50” website: <https://classroom50.org/ubsuny/compphys-fall26/assignments/assignment-2/accept>. This will create a new repository for you on github, titled something like `github.com/ubsuny/ubsuny/compphys-fall26-assignment-2-username`. The repository is located in the `ubsuny` github group, but it is your personal repository, and only you can view it. (The repository is actually not a fork, but rather a brand new repository created by directly copying files from a “template repository.”)

The assignment requires two uploads:

1. **Submit a writeup/lab report to UBLearns.** The writeup should contain your complete solutions to the problems in writing. Feel free to use any format: latex, markdown, etc. For assignments using Jupyter notebooks, you can also do your writeup inside the notebook using Markdown cells, export the notebook to HTML/PDF/etc. (this may require some additional setup), and upload the exported notebook.
2. **Upload your code through Classroom 50.** This just means to `git push` any code you wrote to your GitHub repository. You can push as many times as you like before the due date (in fact, this is a good idea, as it serves as a backup; nothing will be graded before the due date).

Problem 1

25 points

This problem explores the limitations of floating point numbers for a specific physics problem: the electric field of a static dipole. Consider the following configuration, where a charge of $+q$ is placed at $(d/2, 0)$ and a charge $-q$ is placed at $(-d/2, 0)$:

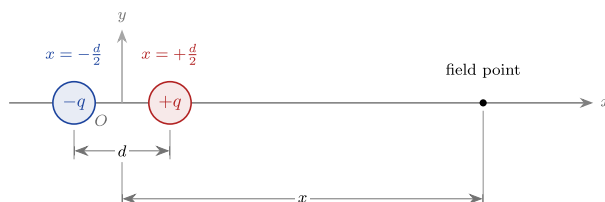


Figure 1: Dipole diagram

At a point along the x -axis, the electric field is given by:

$$E_{\text{exact}}(x) = E_+ - E_- = \frac{q}{(x - d/2)^2} - \frac{q}{(x + d/2)^2}.$$

For $x \gg d$, i.e. far away from the dipole, we can approximate this as:

$$E_{\text{approx}} = \frac{2qd}{x^3}.$$

Note: we are “cheating” a bit by setting $1/(4\pi\epsilon_0) = 1$. This could be arranged through appropriate units, for example. Besides making our lives easier, for numerical computations, it is generally a good idea to keep your numbers around the middle of the range of your data type (e.g., usually close to 1).

Problem 1a

10 points

We will fix $x = 1$ m and $q = 1$, and let d vary. Before programming anything, let’s estimate the values of d where we expect to see floating point problems. Recall that floating point numbers have a fixed number of bits for the mantissa/significand (e.g. 23 bits for a float, 52 bits for a double); any additional significant figures are lost, rounding to the nearest available floating point number. This is potentially a big problem for *subtraction* operations, like $E_{\text{exact}} = E_+ - E_-$: at minimum, we tend to lose significant figures, and in the worse case, the result can be smaller than the floating point error.

You can roughly think of this truncation as an *relative uncertainty*, ϵ . For example, if we restricted ourselves to only 2 significant figures, we can express numbers such as 1.0, 1.1, 1.2, ..., 10., 11., 12., ..., 100., 110., 120., etc. If we try to put 1.0499 into such a float, it will be rounded down to 1.0, which corresponds to a 5% error. Thus, for a 2-digit float, 5% is an upper bound on the the uncertainty introduced by truncation.

In a real C++ program, the `<limits>` library provides `std::numeric_limits<T>::epsilon()`, a templated function that returns an upper bound on the relative uncertainty for type `T` (here,

type `T` is either a `float` or a `double`)¹. For a number `float x`, we can roughly say that `x` has an uncertainty, `dx`, given by `float dx = x * std::numeric_limits<T>::epsilon()`².

Using propagation of uncertainty, estimate the value of d for which $E_{\text{exact}} \approx dE_{\text{exact}}$, for both floats and doubles. Hint: starting from $E_{\text{exact}} = E_+ - E_-$, we have $dE_{\pm} = \epsilon E_{\pm}$. Use propagation of uncertainty to determine dE_{exact} , set $E_{\text{exact}} = dE_{\text{exact}}$ to find the minimum E_{exact} , and then solve for d . Use $\epsilon = 1.19209 \times 10^{-7}$ for floats and $\epsilon = 2.22045 \times 10^{-16}$ for doubles.

In your write-up, please show your work for this calculation as well as the result.

Problem 1b

15 points

Next, let's demonstrate the floating point limit in a real program. The file `problem1/problem1.cc` contains two template functions corresponding to E_{exact} and E_{approx} above. It explicitly instantiates `E_exact` for `float` and `double`, and `E_approx` for `double` (we could also instantiate `E_approx` with `float`, but the results will be very close to `double`, so we won't bother). The example code demonstrates how to compare the three functions for a value of `d=0.01`. The result is:

```
E_exact<double>(0.01)    = 0.020001
E_exact<float>(0.01)     = 0.0200011 Rel. diff. = 5.66858e-06
E_approx<double>(0.01)  = 0.02          Rel. diff. = -4.99994e-05
```

The relative difference (`Rel. diff.`) is calculated using `E_exact<double>` as a reference. Both `E_exact<float>` and `E_approx<double>` are pretty close to `E_exact<double>`, indicating that both are doing a reasonable job for `d=0.01`. However, note that we can already see floating point errors start to creep in.

Modify `problem1.cc` to compare the values of `float E_exact`, `double E_exact`, and `double E_approx` for `d` values of `1.0`, `0.1`, `1.e-2`, `1.e-3`, ..., `1.e-12`. You should see the relative differences become large for certain values of `d`. In your write-up, include the output of the program (a table is fine; a log-scale plot would be a better visualization but we haven't covered that in class yet), and explain any differences between the 3 functions. Does the program match your expectation from Problem 1a?

¹Specifically, `std::numeric_limits<T>::epsilon()` returns the gap between 1.0 and the next highest number.

²Again, this is fairly rough: `epsilon` is a conservative upper bound on what we would actually consider the uncertainty, but we are just doing an order-of-magnitude estimate.

Problem 2

25 points

The **standard deviation**, σ , of a set of N measurements, $\{x_0, x_1, \dots, x_{N-1}\}$, is given by:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=0}^{N-1} (x_i - \langle x \rangle)^2$$

where $\langle x \rangle$ is the mean of the $\{x_i\}$. Naively, implementing this formula in a program requires 2 **for** loops: one to compute the mean $\langle x \rangle$, and a second to compute the outer sum.

In `problem2/problem2.cc`, we define a base and derived class:

1. Base class `StdDevBase` is a pure virtual class. It defines the interface for a standard deviation calculator, but not the implementation. It provides a data loader common method, `LoadData(std::vector<float> data_in)`, for the derived classes, and specifies a virtual function, `virtual float get_stddev()=0`, to indicate that derived classes should implement this function (i.e. override).
2. Derived class `StdDev` implements the standard deviation formula, `float get_stddev()`, using the naive “2-pass” method. The first pass computes the mean, $\langle x \rangle = \frac{1}{N} \sum_{i=0}^{N-1} x_i$, and the second pass computes $\sum_{i=0}^{N-1} (x_i - \langle x \rangle)^2$.

We also provide a function, `std::vector<float> GenerateData(unsigned int n, float mean, float sigma)`, to create a vector of random, gaussian-distributed numbers (don’t worry about the details, we will cover this much later in the course).

Finally, `int main()` function calls `GenerateData()`, then uses an instance of `StdDev` to compute the standard deviation. Since the random numbers are drawn from a gaussian distribution, the computed standard deviation should be nearly equal to the gaussian `sigma`.

Problem 2a

15 points

There is a classic “trick” to compute the standard deviation in one pass³. With $\langle x \rangle = \sum_{i=0}^{N-1} x_i / N$, we simply rearrange the formula as follows:

$$\begin{aligned} \sigma^2 &= \frac{1}{N-1} \sum_{i=0}^{N-1} (x_i^2 - 2x_i\langle x \rangle + \langle x \rangle^2) \\ &= \frac{1}{N-1} \left[\left(\sum_{i=0}^{N-1} x_i^2 \right) - 2 \left(\sum_{i=0}^{N-1} x_i \right) \langle x \rangle + \left(\sum_{i=0}^{N-1} \langle x \rangle^2 \right) \right] \\ &= \frac{1}{N-1} \left[\left(\sum_{i=0}^{N-1} x_i^2 \right) - \left(\sum_{i=0}^{N-1} x_i \right)^2 / N \right] \end{aligned}$$

With this formula, we can compute the standard deviation with only one **for** loop, in which we accumulate the sums for both $\sum_{i=0}^{N-1} x_i^2$ and $\sum_{i=0}^{N-1} x_i$.

³You might be familiar with the simpler formula for the biased standard deviation, $\sigma^2 = \frac{1}{N} (\langle x^2 \rangle - \langle x \rangle^2)$.

Implement the one-pass formula in a new derived class, `StdDevFast`, which derives from `StdDevBase`. You can start by copying the code for the `StdDev`, and just modify the `get_stddev()` function.

In `int main()`, add a few lines to use the `StdDevFast` class to compute the standard deviation of the same vector of random numbers. In your write-up, just copy the output of the program, showing the standard deviation computed using the two derived classes.

Problem 2b

10 points

Using the `StdDevFast` class from Problem 2a, let's try to quantify the performance difference.

Adapt the code from the example `CompPhys/ReviewCpp/BasicExamples/func_timing.cpp` to measure the performance of the two classes. You should increase the size of the vector to observe a difference, if needed. Is the `StdDevFast` class actually faster?

Problem 2c

10 points

PHY505 students only

The standard deviation computation involves a subtraction, so we should once again be wary of floating point limitations. Further, the two classes implement the subtraction rather differently: the `StdDevFast` class only performs the subtraction once, but with potentially very large numbers (e.g. $\sum_{i=0}^{N-1} x_i^2$) with a small difference, exactly the case we need to worry about.

Find some values of N and the gaussian mean and width where the standard deviation computation breaks down. You should be able to:

1. Find a case where `StdDev` works, but `StdDevFast` fails.
 2. Find a case where both `StdDev` and `StdDevFast` fail.
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